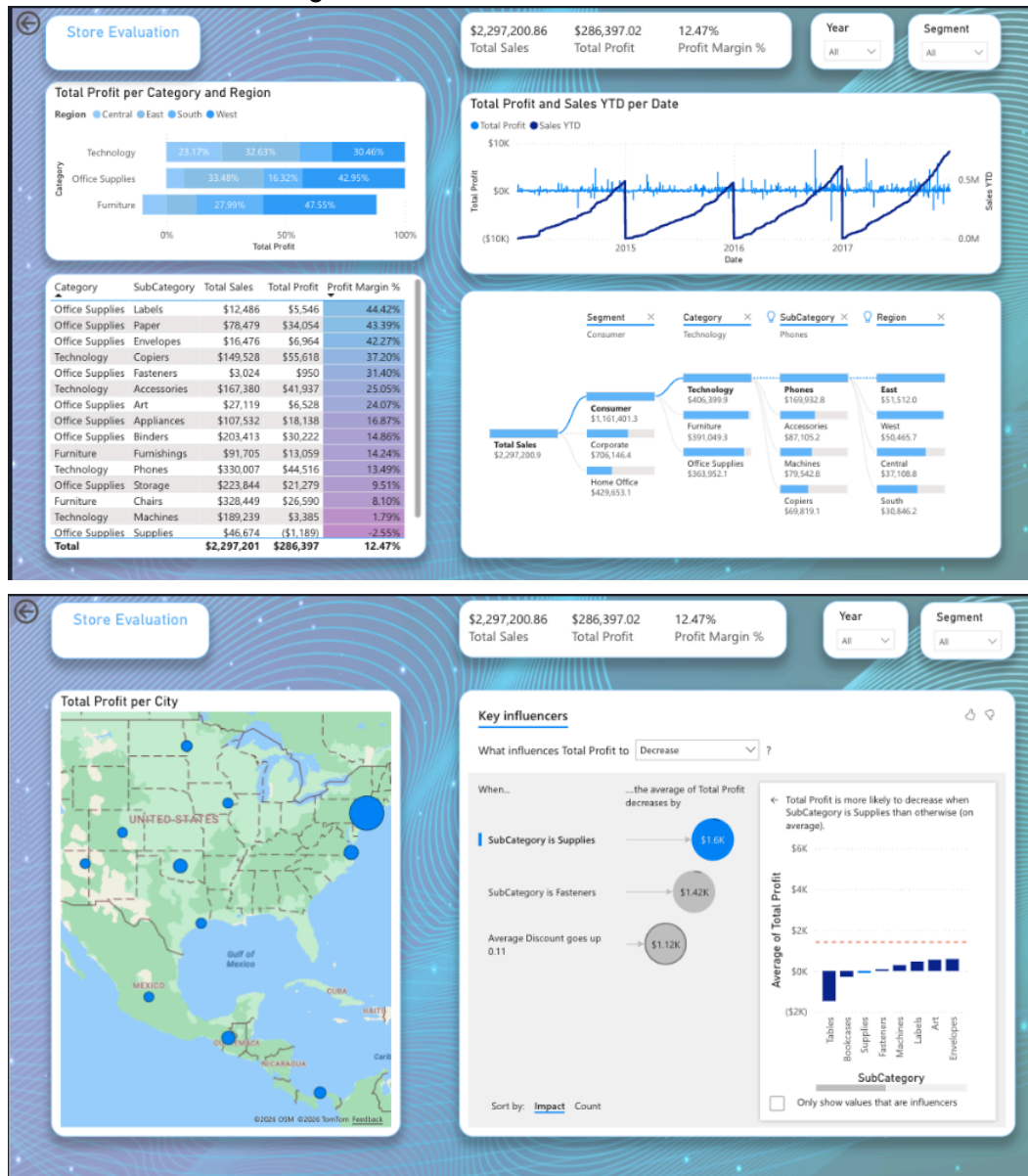


BUSINESS DATA ANALYSIS

REPORT

Project Objective

The aim of the project was to conduct an analysis of sales data for a trading company using Microsoft Power BI and to design an interactive report supporting the assessment of sales performance and profitability. The project was intended to provide a practical application of data analysis techniques covered during laboratory classes, such as data transformation, data model construction, creating measures in the DAX language, visualization design, and the use of artificial intelligence features.



Project Assumptions

The project was carried out based on the following assumptions:

- the analysis covers sales data including orders, products, customers, regions, and time,
- the data was subjected to a transformation and cleaning process in Power Query,
- the data model was designed as a star schema with one fact table and dimension tables,

- the project uses calculated measures and calculated columns written in the DAX language,
- the report includes interactive visualizations enabling data analysis across various criteria,
- the project makes use of artificial intelligence features available in Power BI to identify factors influencing financial results.

Source Data

The project used the publicly available

The project used the publicly available **Superstore Dataset**, sourced from Kaggle.

The data contains information on:

- sales orders,
- products (categories and subcategories),
- customers and customer segments,
- locations (cities, regions),
- order and shipping dates,
- sales values, profit, quantity, and discounts.

Link: <https://www.kaggle.com/datasets/vivek468/superstore-dataset-final/data>

The dataset was imported into Power BI in CSV format and then processed further.

Steps Performed in Power BI with Screenshots

1. Power Query (Data Transformation)

1.1 Checking and Correcting Data Types, Removing Missing Data and Errors

In the first stage, data types were corrected to ensure the accuracy of numerical calculations and time-based analyses.

Data type changes:

- Row ID → integer
- Order Date, Ship Date → type changed with regional settings (data originates from the USA, so dates were in month/day/year format) — changed to date in day/month/year format
- Postal Code, Quantity → changed to integer
- Sales, Profit, Discount → changed to decimal numbers with regional settings (in the USA a period is used instead of a comma as the decimal separator)

During changes in Power Query:

1.2 Sales	1.2 Quantity	% Discount	1.2 Profit	WŁAŚCIWOŚCI
261,96	2	0,00%	41,9136	Nazwa
731,94	3	0,00%	219,582	Sample - Superstore
14,62	2	0,00%	6,8714	Wszystkie właściwości
957,5775	5	45,00%	-383,031	ZASTOSOWANE KROKI
22,368	2	20,00%	2,5164	Źródło
48,86	7	0,00%	14,1694	Nagłówki o podwyższonym poziomie
7,28	4	0,00%	1,9656	Przefiltrowano wiersze
907,152	6	20,00%	90,7152	Zmieniono typ z ustawieniami regionalnymi
18,504	3	20,00%	5,7825	Zmieniono typ z ustawieniami regionalnymi1
114,9	5	0,00%	34,47	Zmieniono typ z ustawieniami regionalnymi2
1706,184	9	20,00%	85,3092	Zmieniono typ
911,424	4	20,00%	68,3568	Zmieniono typ z ustawieniami regionalnymi3
15,552	3	20,00%	5,4432	Zmieniono typ z ustawieniami regionalnymi4
407,976	3	20,00%	132,5922	X Zmieniono typ1
68,81	5	80,00%	-123,858	
2,544	3	80,00%	-3,816	
665,88	6	0,00%	13,3176	
55,5	2	0,00%	9,99	
8,56	2	0,00%	2,4824	
213,48	3	20,00%	16,011	
22,72	4	20,00%	7,384	
19,46	7	0,00%	5,0596	

Postal Code	1.2 Row ID	Order ID	Order Date	Ship Date
42420	1	CA-2016-152156	08.11.2016	11.11.2016
42420	2	CA-2016-152156	08.11.2016	11.11.2016
90036	3	CA-2016-138688	12.06.2016	16.06.2016
33311	4	US-2015-108966	11.10.2015	18.10.2015
33311	5	US-2015-108966	11.10.2015	18.10.2015
90032	6	CA-2014-115812	09.06.2014	14.06.2014
90032	7	CA-2014-115812	09.06.2014	14.06.2014
90032	8	CA-2014-115812	09.06.2014	14.06.2014
90032	9	CA-2014-115812	09.06.2014	14.06.2014
90032	10	CA-2014-115812	09.06.2014	14.06.2014
90032	11	CA-2014-115812	09.06.2014	14.06.2014
90032	12	CA-2014-115812	09.06.2014	14.06.2014
90032	13	CA-2017-114412	15.04.2017	20.04.2017
90032	14	CA-2016-161389	05.12.2016	10.12.2016
90032	15	US-2015-118983	22.11.2015	26.11.2015
90032	16	US-2015-118983	22.11.2015	26.11.2015
28027	17	CA-2014-105893	11.11.2014	18.11.2014
98102	18	CA-2014-167164	13.05.2014	15.05.2014

Records with missing or erroneous values that could distort the analysis results were checked. No such records were found in this dataset.

1.2 Standardising Column Names

Selective standardisation of column names was applied. Names containing spaces and those that would be used in DAX calculations were renamed, while technical identifiers were left in their original form to maintain consistency with the data source.

Order Date	→ OrderDate
Ship Date	→ ShipDate
Ship Mode	→ ShipMode
Customer Name	→ CustomerName
Postal Code	→ PostalCode
Sub-Category	→ SubCategory
Product Name	→ ProductName

Final result:

OrderDate	ShipDate	ShipMode	Customer ID	CustomerName
PostalCode	Region	Product ID	Category	SubCategory
ProductName				

1.3 Splitting Text Columns

In the Power Query editor, a split column operation was applied using a delimiter. The column containing the customer's full name was split by space into two separate columns, enabling data normalisation and improving model readability.

Dzielenie kolumny według ogranicznika

Określ ogranicznik służący do dzielenia kolumny tekstowej.

Wybierz lub wprowadź ogranicznik

Spacja

Podziel przy

☒ Ogranicznik najdalej z lewej strony

☐ Ogranicznik najdalej z prawej strony

☐ Każde wystąpienie ogranicznika

> Opcje zaawansowane

Znak cudzysłowu

"

☐ Podziel przy użyciu znaków specjalnych

Wstaw znak specjalny

OK

Anuluj

As a result, two columns are created:

- CustomerName.1 → renamed to CustomerFirstName
- CustomerName.2 → renamed to CustomerLastName

CustomerFirstName	CustomerLastName
Claire	Gute
Claire	Gute
Darrin	Van Huff
Sean	O'Donnell
Sean	O'Donnell
Brosina	Hoffman
Brosina	Hoffman
Brosina	Hoffman
Brosina	Hoffman
Brosina	Hoffman
Brosina	Hoffman
Andrew	Allen
Irene	Maddox
Harold	Pawlan

For compound surnames such as Van Buren, the project adopts a simplification of splitting at the first space, which is sufficient for the purposes of aggregated analysis.

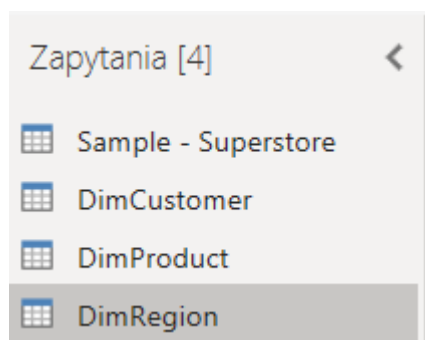
1.3 Adding Helper Columns from Date (OrderDate)

Based on the order date, helper columns were added to enable data analysis by year, month, and quarter.

123 Rok	123 Miesiąc	A ^B _C Nazwa miesiąca	123 Kwartał
2016	11	listopad	4
2016	11	listopad	4
2016	6	czerwiec	2
2015	10	październik	4
2015	10	październik	4
2014	6	czerwiec	2
2014	6	czerwiec	2
2014	6	czerwiec	2
2014	6	czerwiec	2
2014	6	czerwiec	2
2014	6	czerwiec	2
2014	6	czerwiec	2

1.4 Duplicating Tables — Preparing Dimensions

To prepare the data model, separate dimension tables were created by duplicating the source table.



1.5 Creating Dimension Tables (Data Cleaning)

DimCustomer consists of:

- Customer ID (duplicates removed)
- CustomerFirstName
- CustomerLastName
- Segment

Zapytania [4]

×

✓

f_x

= Table.Distinct(#"Usunięto kolumny", {"Customer ID"})

Customer ID

A^B_C

CustomerFirstName

A^B_C

CustomerLastName

A^B_C

Segment

1

CG-12520

Claire

Gute

Consumer

2

DV-13045

Darrin

Van Huff

Corporate

3

SO-20335

Sean

O'Donnell

Consumer

4

BH-11710

Brosina

Hoffman

Consumer

5

AA-10480

Andrew

Allen

Consumer

6

IM-15070

Irene

Maddox

Consumer

Sample - Superstore

DimCustomer

DimProduct

DimRegion

A customer dimension table was created containing unique customer records and their segmentation.

DimProduct consists of:

- Product ID (duplicates removed)
- ProductName
- Category
- SubCategory

Zapytania [4] fx = Table.Distinct(#"Usunięto inne kolumny", {"Product ID"})

	Product ID	ProductName	Category	SubCategory
1	FUR-BO-10001798	Bush Somerset Collection Bookcase	Furniture	Bookcases
2	FUR-CH-10000454	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	Furniture	Chairs
3	OFF-LA-10000240	Self-Adhesive Address Labels for Typewriters by Universal	Office Supplies	Labels
4	FUR-TA-10000577	Bretford CR4500 Series Slim Rectangular Table	Furniture	Tables
5	OFF-ST-10000760	Eldon Fold 'N Roll Cart System	Office Supplies	Storage
6	FUR-FU-10001487	Eldon Expressions Wood and Plastic Desk Accessories, Cherry Wood	Furniture	Furnishings
7	OFF-AR-10002833	Newell 322	Office Supplies	Art
8	TEC-PH-10002275	Mitel 5320 IP Phone VoIP phone	Technology	Phones

A product dimension table was created containing information on product categories and subcategories.

DimRegion consists of:

- Region
- State
- City

Duplicates were removed across all three columns.

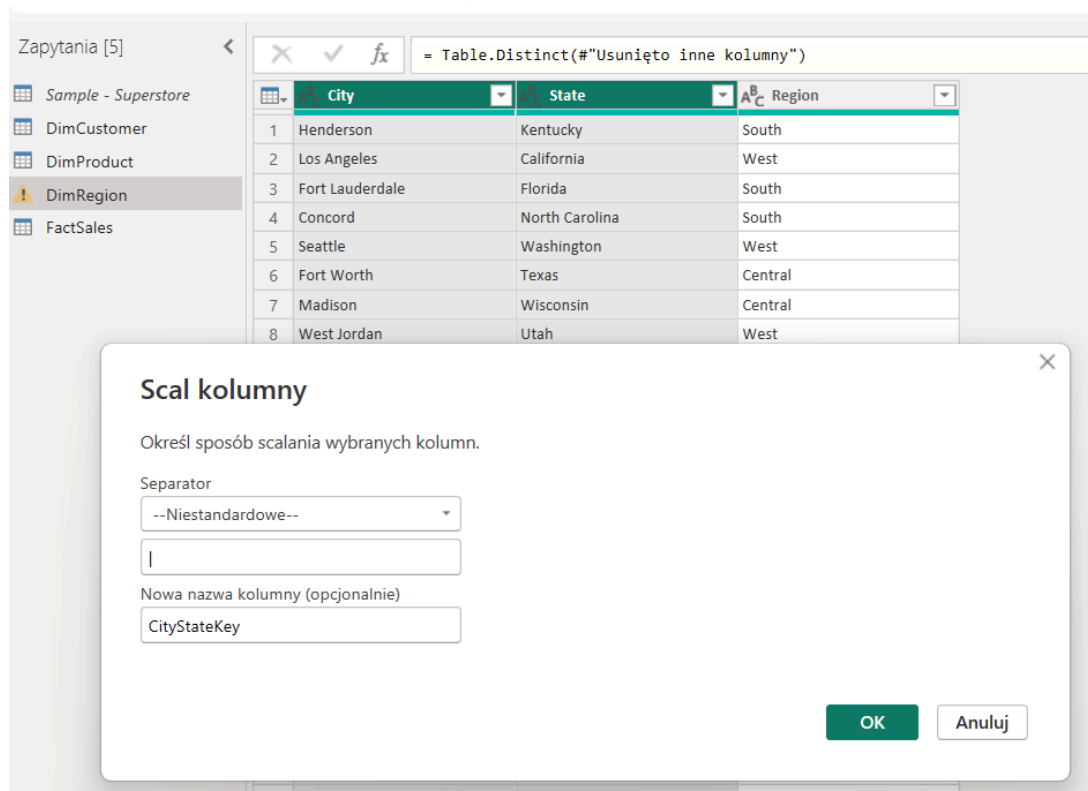
Zapytania [4] fx = Table.Distinct(#"Usunięto inne kolumny")

	City	State	Region
1	Henderson	Kentucky	South
2	Los Angeles	California	West
3	Fort Lauderdale	Florida	South
4	Concord	North Carolina	South
5	Seattle	Washington	West
6	Fort Worth	Texas	Central
7	Madison	Wisconsin	Central
8	West Jordan	Utah	West

A regional dimension table was created to enable geographical sales analysis.

To ensure uniqueness of the key in the region dimension table, a composite key was created based on city and state. The relationship between the fact table and the dimension table was defined using this key, which preserved a one-to-many relationship and compliance with the star schema. Since city alone is not unique, a composite City–State key was used.

Creating the composite key (CityState):



A character separator not present in the source data was used to create the composite key. This allows unambiguous distinction of key components and avoids potential data collisions. Duplicates were then removed on the new key.

1.6 Preparing the Fact Table — FactSales

The main table "Sample – Superstore" was retained as the source/ETL reference, while FactSales was created for analysis — without Year/Quarter/etc. columns (time will be in DimDate).

Zapytania [5]

= Table.RemoveColumns(#"Wstawiono kwartał",{"ShipMode", "CustomerFirstName", "CustomerLastName", "Segment", "Country", ...})

Row ID	Order ID	OrderDate	ShipDate	Customer ID	City
1	CA-2016-152156	08.11.2016	11.11.2016	CG-12520	Henderson
2	CA-2016-152156	08.11.2016	11.11.2016	CG-12520	Henderson
3	CA-2016-138688	12.06.2016	16.06.2016	DV-13045	Los Angeles
4	US-2015-108966	11.10.2015	18.10.2015	SO-20335	Fort Lauderdale
5	US-2015-108966	11.10.2015	18.10.2015	SO-20335	Fort Lauderdale
6	CA-2014-115812	09.06.2014	14.06.2014	BH-11710	Los Angeles
7	CA-2014-115812	09.06.2014	14.06.2014	BH-11710	Los Angeles

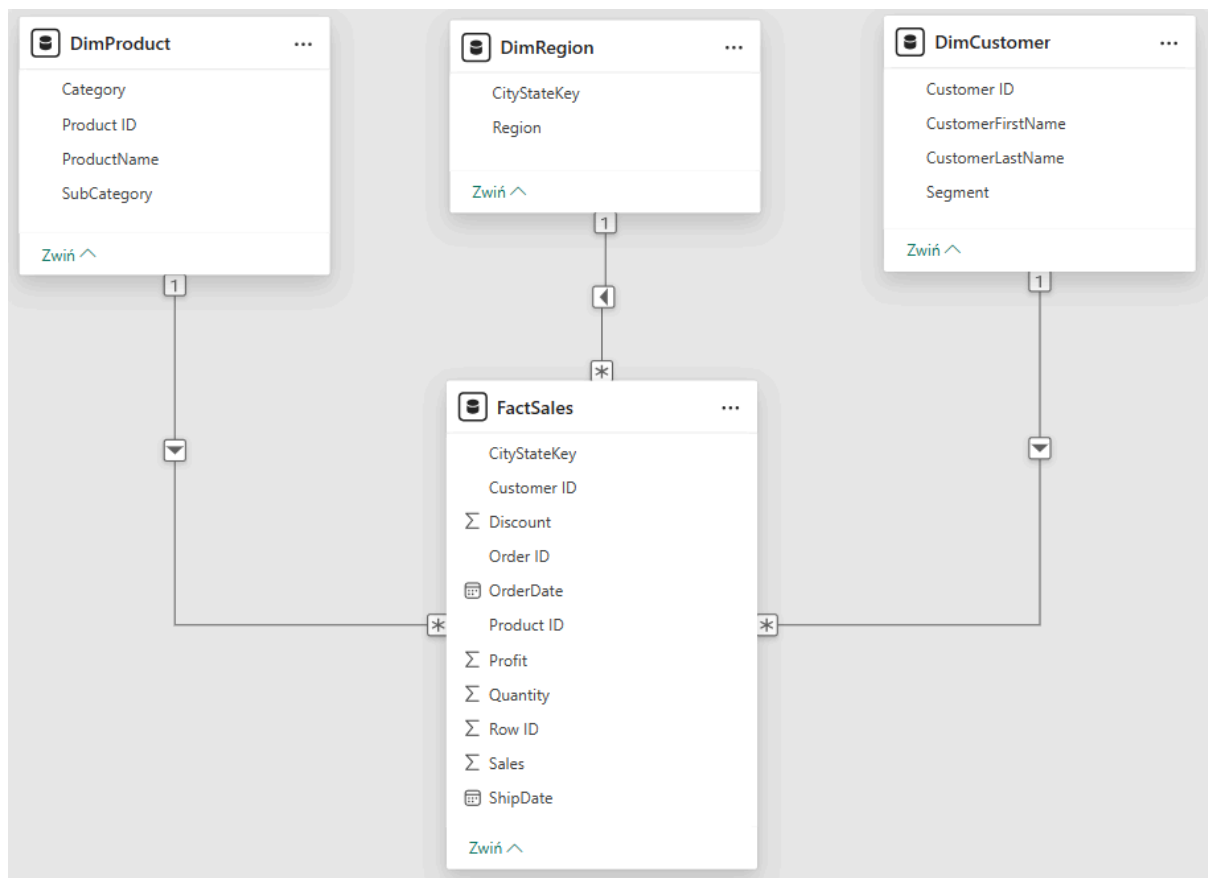
The FactSales fact table was created by duplicating the source table and removing descriptive and helper columns. The fact table retains measures (Sales, Profit, Quantity, Discount) as well as keys and fields identifying the transaction, in line with star schema principles.

2. Data Model

2.1 Relationships

A star schema was applied with one fact table and three dimension tables. Relationships are one-to-many, which enables correct analyses and model scalability.

The central table is FactSales containing quantitative measures, while DimProduct, DimCustomer, and DimRegion provide analytical context. Relationships between tables are one-to-many and unidirectional, which ensures correct operation of DAX measures and high model readability.



2.2 Date Table (DimDate in DAX)

The DimDate table was marked as a date table, which enabled correct operation of DAX time intelligence functions such as TOTALYTD and year-over-year comparisons.

Modelowanie

Narzędzia tabel

DimDate

```

1 DimDate =
2 ADDCOLUMNS (
3     CALENDAR (
4         MIN ( FactSales[OrderDate] ),
5         MAX ( FactSales[OrderDate] )
6     ),
7     "Rok", YEAR ( [Date] ),
8     "NumerMiesiaca", MONTH ( [Date] ),
9     "Miesiac", FORMAT ( [Date], "MMMM" ),
10    "RokMiesiac", FORMAT ( [Date], "YYYY-MM" ),
11    "Kwartal", "Q" & FORMAT ( [Date], "Q" )
12 )

```

Oznacz jako tabelę dat

Aby umożliwić tworzenie wizualizacji, tabeli i szybkich miar związanych z datą przy użyciu tabeli daty.

Pamiętaj, że wszelkie wbudowane tabele dat, które są już skojarzone z tą tabelą, zostaną u języka DAX odwołujące się do nich mogą zostać uszkodzone. [Dowiedz się więcej](#)

Oznacz jako tabelę dat

☒ Włączony

Wybierz kolumnę daty

Date

Walidacja zakończona pomyślnie

2.3 Creating Measures

Narzędzia miar

Nazwa Miara

Format \$

Kategoria danych Bez kategorii

Struktura

Formatowanie

Właściwości

Wizualizacje

Utwórz element wizualny

Filtr

Utwórz wizualizację na podstawie danych

Zaznacz lub przeciągnij pola z okienka Dane na kanwę raportu.

Dane

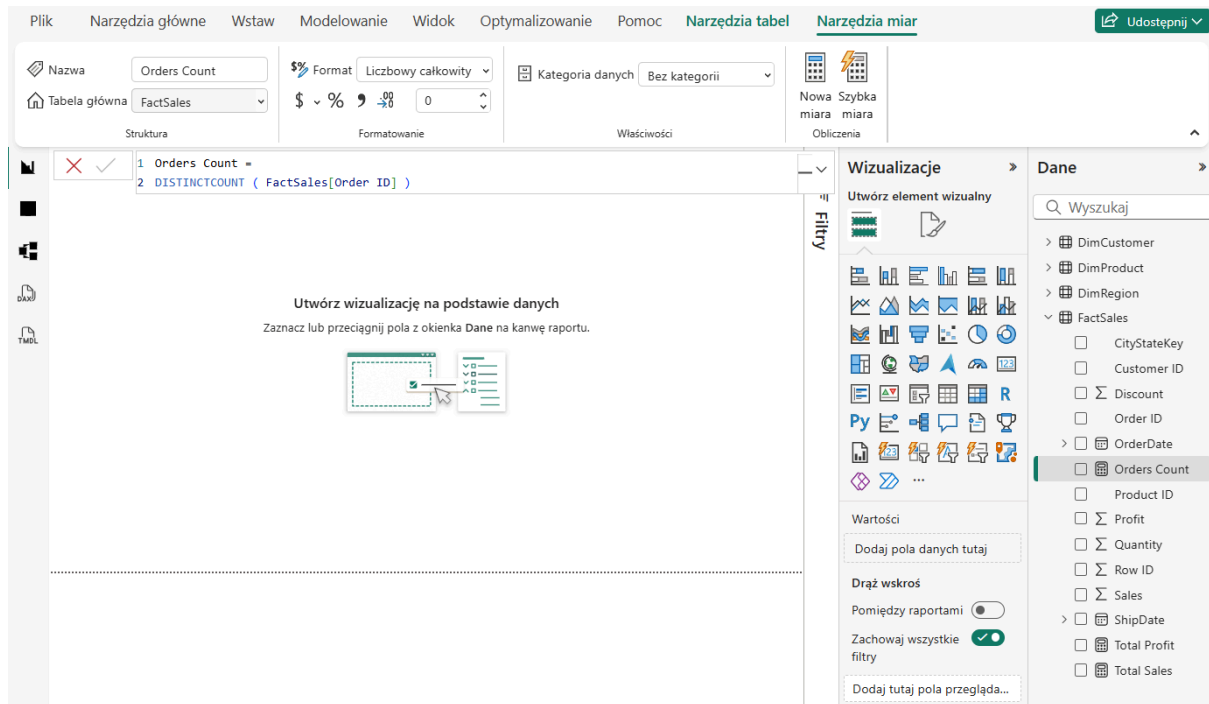
Wyszukaj

- DimCustomer
- DimProduct
- DimRegion
- FactSales
 - CityStateKey
 - Customer ID
 - Discount
 - Miara
 - Order ID
 - Product ID
 - Profit
 - Quantity
 - Row ID
 - Sales
 - ShipDate

Basic Measures:

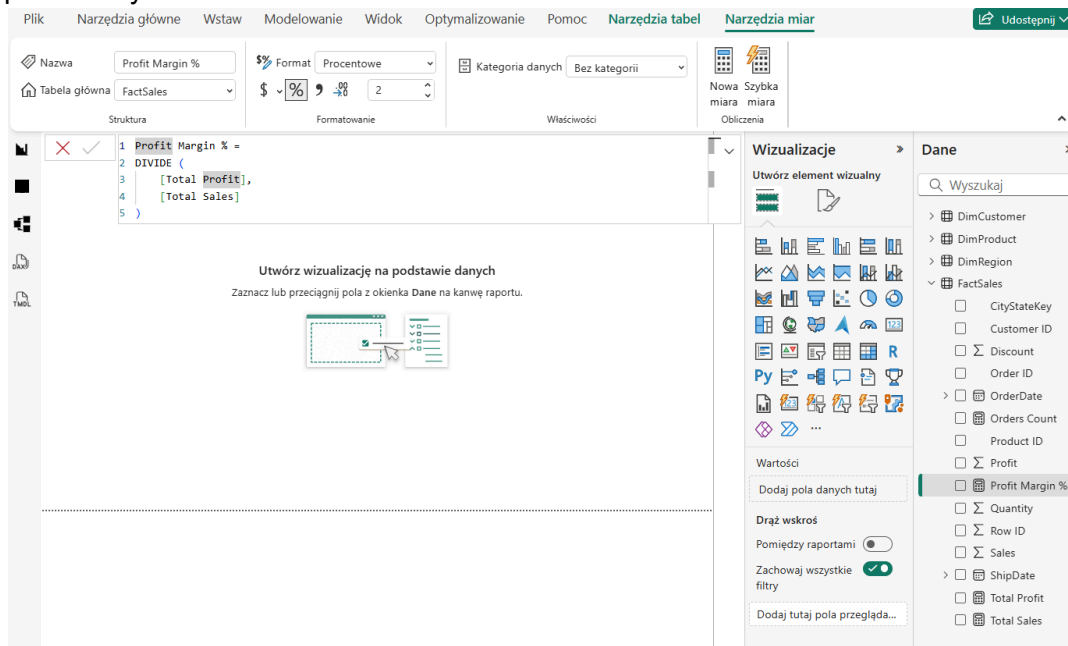
Basic DAX measures were created to enable analysis of sales, profit, and number of orders.

Total Sales:



Profitability:

Profit Margin — a profit margin measure was defined as the ratio of profit to sales, enabling profitability assessment.



Time Intelligence:

DAX time intelligence functions were applied, enabling year-over-year comparisons and year-to-date cumulative sales analysis.

Sales YTD (Year-to-Date):

Plik Narzędzia główne Wstaw Modelowanie Widok Optymalizowanie Pomoc **Narzędzia tabel** **Narzędzia miar**

Nazwa: Sales YTD Format: Ogólne Tabela główna: FactSales

Struktura: 1 Sales YTD =
2 TOTALYTD (
3 [Total Sales],
4 DimDate[Date]
5)

Formatowanie: \$ % % Autom...

Właściwości: Własności

Nowa Szybka miara miara Obliczenia

Utwórz wizualizację na podstawie danych
Zaznacz lub przeciągnij pola z okienka Dane na kanwę raportu.

Dane: Wyszukaj

- DimCustomer
- DimDate
- DimProduct
- DimRegion
- FactSales
 - CityStateKey
 - Customer ID
 - Discount
 - Order ID
 - OrderDate
 - Orders Count
 - Product ID
 - Profit
 - Profit Margin %
 - Quantity
 - Row ID
 - Sales
 - Sales YTD**
 - ShipDate
 - Total Profit
 - Total Sales

Sales Year-over-Year (YoY — difference):

Plik Narzędzia główne Wstaw Modelowanie Widok Optymalizowanie Pomoc **Narzędzia tabel** **Narzędzia miar**

Nazwa: Sales YoY Format: Ogólne Tabela główna: FactSales

Struktura: 1 Sales YoY =
2 [Total Sales]
3 - CALCULATE (
4 [Total Sales],
5 SAMEPERIODLASTYEAR (DimDate[Date])
6)

Formatowanie: \$ % % Autom...

Właściwości: Własności

Nowa Szybka miara miara Obliczenia

Utwórz wizualizację na podstawie danych
Zaznacz lub przeciągnij pola z okienka Dane na kanwę raportu.

Dane: Wyszukaj

- DimCustomer
- DimDate
- DimProduct
- DimRegion
- FactSales
 - CityStateKey
 - Customer ID
 - Discount
 - Order ID
 - OrderDate
 - Orders Count
 - Product ID
 - Profit
 - Profit Margin %
 - Quantity
 - Row ID
 - Sales
 - Sales YoY**
 - Sales YTD
 - ShipDate
 - Total Profit
 - Total Sales

Sales Year-over-Year (%):

Plik Narzędzia główne Wstaw Modelowanie Widok Optymalizowanie Pomoc **Narzędzia tabel** **Narzędzia miar**

Nazwa: Sales YoY % Format: Procentowe Tabela główna: FactSales

Struktura: `1 Sales YoY % =
2 DIVIDE (
3 [Sales YoY],
4 CALCULATE (
5 [Total Sales],
6 SAMEPERIODLASTYEAR (DimDate[Date])
7)
8)`

Formatowanie: \$ % 9 -00 2

Właściwości: Bez kategorii

Nowa Szybka miara miara Obliczenia

Utwórz wizualizację na podstawie danych
Zaznacz lub przeciągnij pola z okienka Dane na kanwę raportu.

Dane

Wyszukaj

- DimCustomer
- DimDate
- DimProduct
- DimRegion
- FactSales
 - CityStateKey
 - Customer ID
 - Discount
 - Order ID
 - OrderDate
 - Orders Count
 - Product ID
 - Profit
 - Profit Margin %
 - Quantity
 - Row ID
 - Sales
 - Sales YoY
 - Sales YoY %**
 - Sales YTD
 - ShipDate
 - Total Profit
 - Total Sales

Discounts:

Measures were created to analyse the impact of discounts on sales.

Average Discount:

Plik Narzędzia główne Wstaw Modelowanie Widok Optymalizowanie Pomoc **Narzędzia tabel** **Narzędzia miar**

Nazwa: Average Discount Format: Procentowe Tabela główna: FactSales

Struktura: `1 Average Discount =
2 AVERAGE (FactSales[Discount])`

Formatowanie: \$ % 9 -00 2

Właściwości: Bez kategorii

Nowa Szybka miara miara Obliczenia

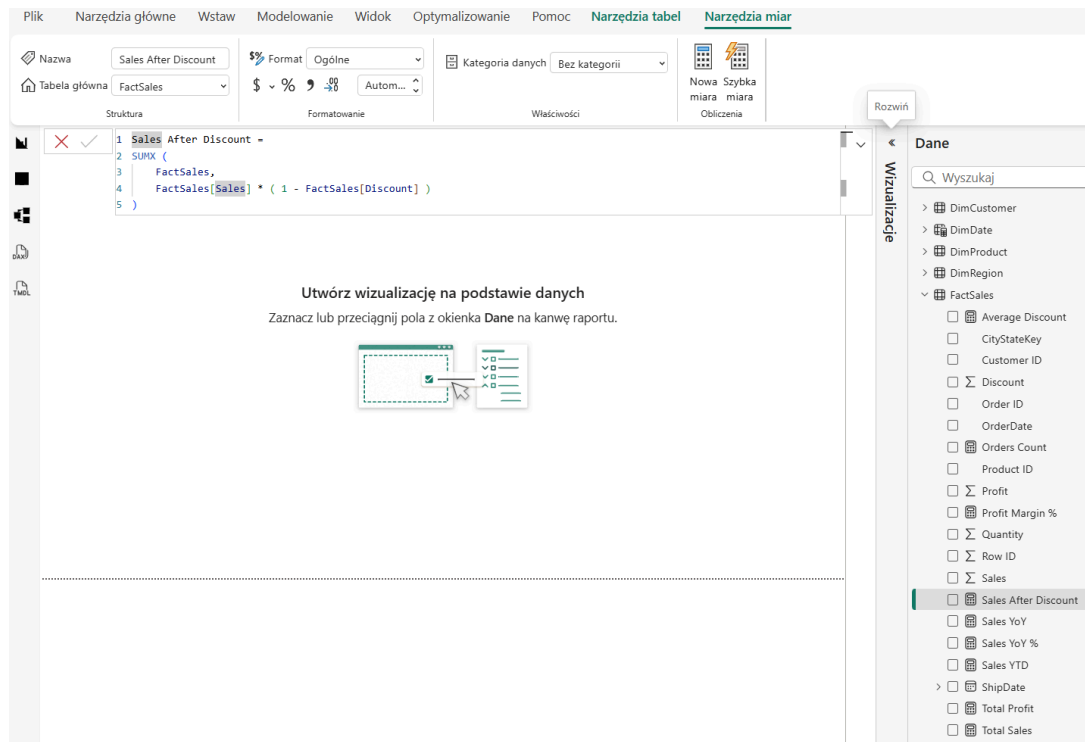
Utwórz wizualizację na podstawie danych
Zaznacz lub przeciągnij pola z okienka Dane na kanwę raportu.

Dane

Wyszukaj

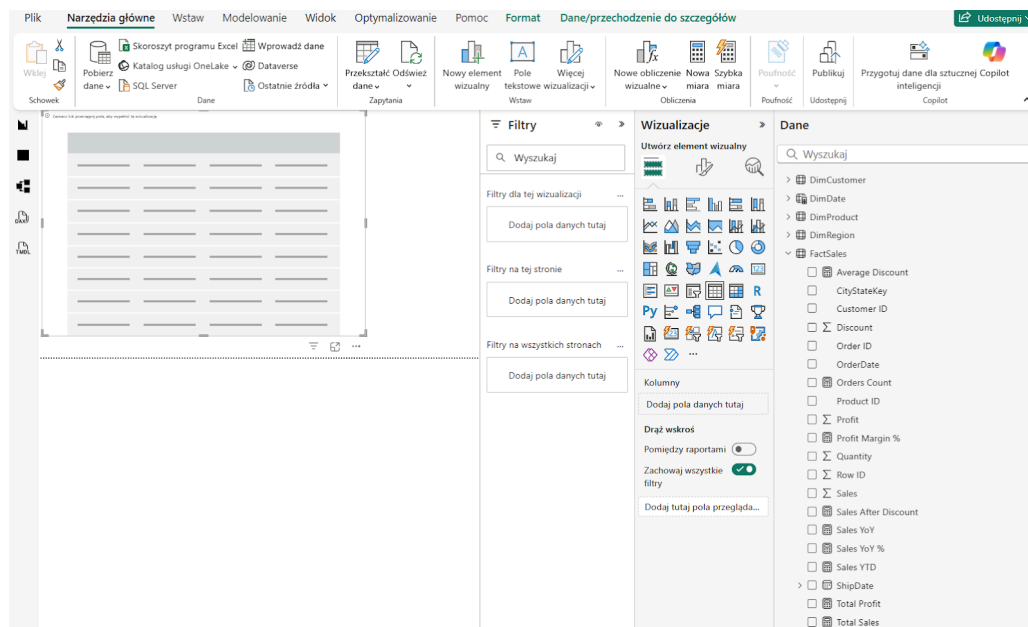
- DimCustomer
- DimDate
- DimProduct
- DimRegion
- FactSales
 - Average Discount**
 - CityStateKey
 - Customer ID
 - Discount
 - Order ID
 - OrderDate
 - Orders Count
 - Product ID
 - Profit
 - Profit Margin %
 - Quantity
 - Row ID
 - Sales
 - Sales YoY
 - Sales YoY %
 - Sales YTD
 - ShipDate
 - Total Profit
 - Total Sales

Sales After Discount:



Measure Testing

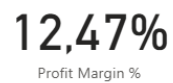
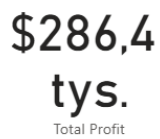
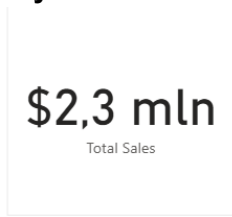
To verify the correctness of the data model and DAX measures, tests were conducted using tables and slicers. The response of measures to product, customer, and time filters was checked. Correct behaviour confirmed the proper configuration of relationships and analytical measures.



3. Created Visualizations — Report View

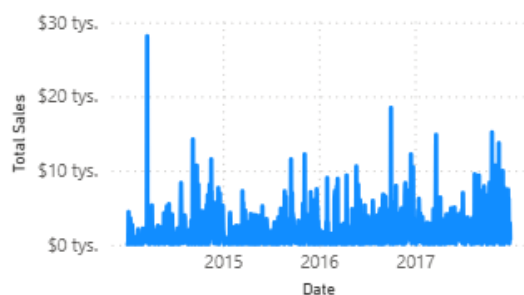
The process of building the final visualization is described below.

Key Performance Indicators (KPIs) showing sales, profit, and profitability:



Sales over time:

Total Sales wg Date

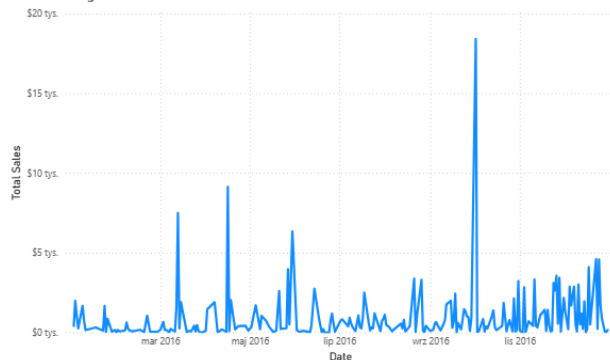


\$207,11
tys.
Total Sales

\$31 tys.
Total Profit

14,97%
Profit Margin %

Total Sales wg Date



Rok
☐ 2014
☐ 2015
☒ 2016
☐ 2017

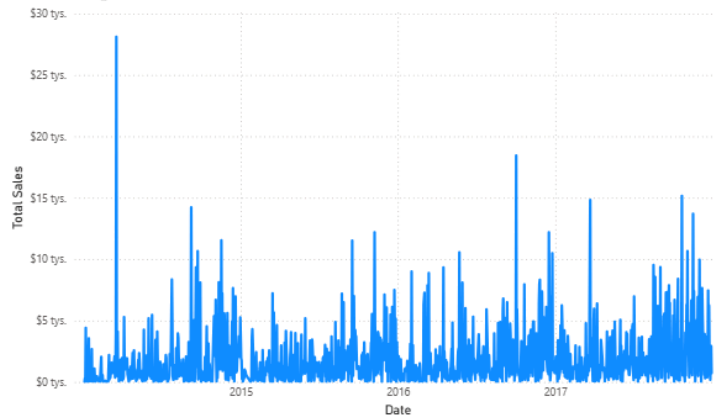
Segment
☐ Consumer
☒ Corporate
☐ Home Office

\$2,3 mln
Total Sales

\$286,4
tys.
Total Profit

12,47%
Profit Margin %

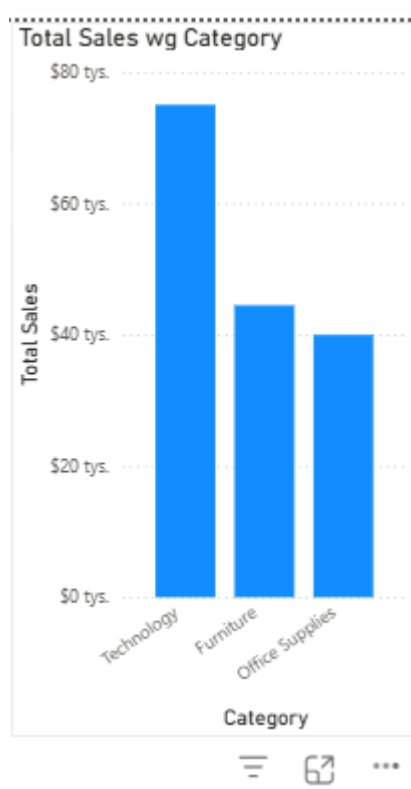
Total Sales wg Date



Rok
☐ Zaznacz wszystkie
☐ 2015
☐ 2017
☐ 2014
☐ 2016

Segment
☐ Zaznacz wszystkie
☐ Corporate
☐ Consumer
☐ Home Office

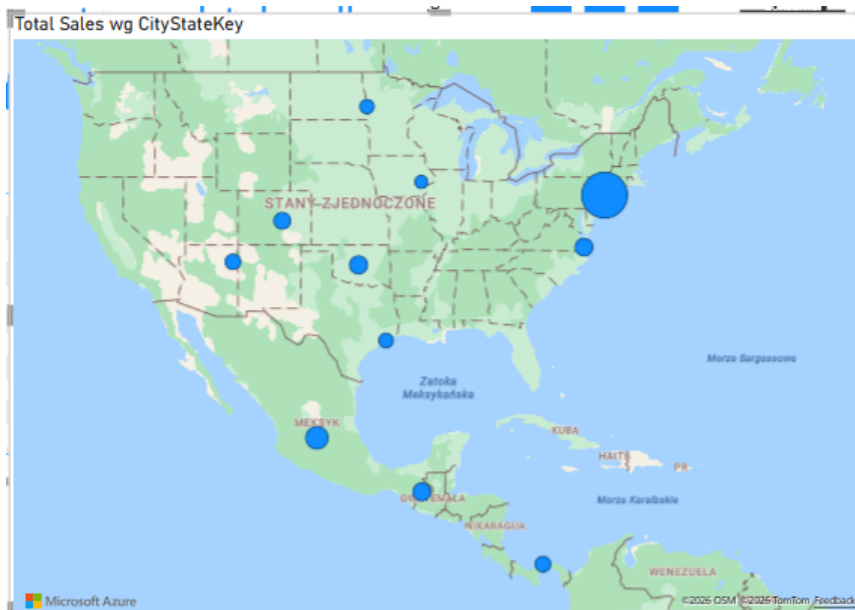
Sales by Category:



Product Profitability:

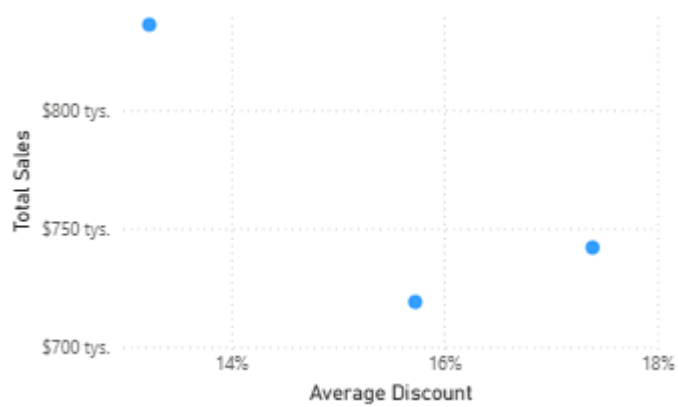
Category	SubCategory	Total Sales	Total Profit	Profit Margin %
Technology	Accessories	\$167 380,318	\$41 936,6357	25,05%
Office Supplies	Appliances	\$107 532,161	\$18 138,0054	16,87%
Office Supplies	Art	\$27 118,792	\$6 527,787	24,07%
Office Supplies	Binders	\$203 412,733	\$30 221,7633	14,86%
Furniture	Bookcases	\$114 879,9963	(\$3 472,556)	-3,02%
Furniture	Chairs	\$328 449,103	\$26 590,1663	8,10%
Technology	Copiers	\$149 528,03	\$55 617,8249	37,20%
Office Supplies	Envelopes	\$16 476,402	\$6 964,1767	42,27%
Office Supplies	Fasteners	\$3 024,28	\$949,5182	31,40%
Furniture	Furnishings	\$91 705,164	\$13 059,1436	14,24%
Office Supplies	Labels	\$12 486,312	\$5 546,254	44,42%
Technology	Machines	\$189 238,631	\$3 384,7569	1,79%
Office Supplies	Paper	\$78 479,206	\$34 053,5693	43,39%
Technology	Phones	\$330 007,054	\$44 515,7306	13,49%
Office Supplies	Storage	\$223 843,608	\$21 278,8264	9,51%
Office Supplies	Supplies	\$46 673,538	(\$1 189,0995)	-2,55%
Furniture	Tables	\$206 965,532	(\$17 725,4811)	-8,56%
Suma		\$2 297 200,8603	\$286 397,0217	12,47%

Sales by Location:



Impact of Discounts on Sales:

Average Discount i Total Sales wg Category



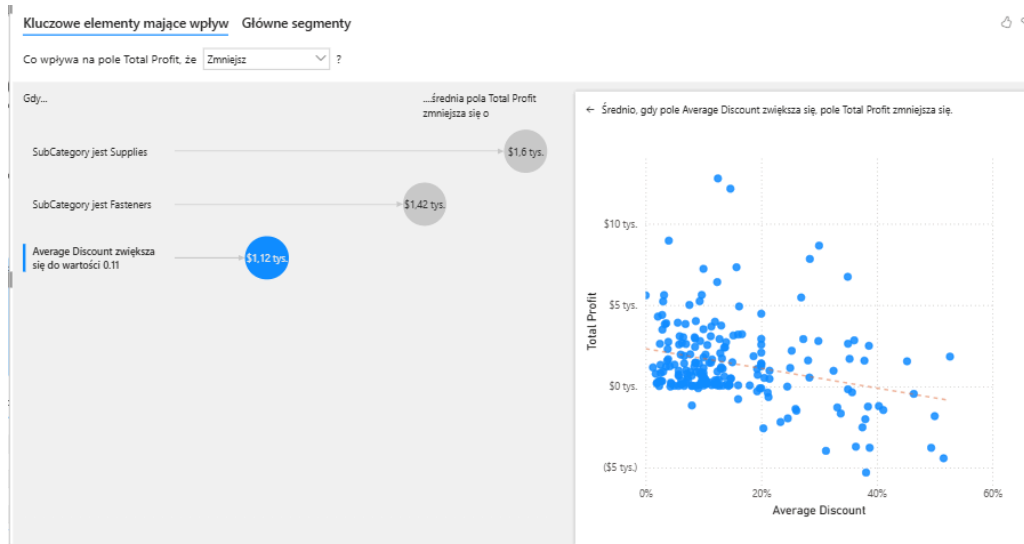
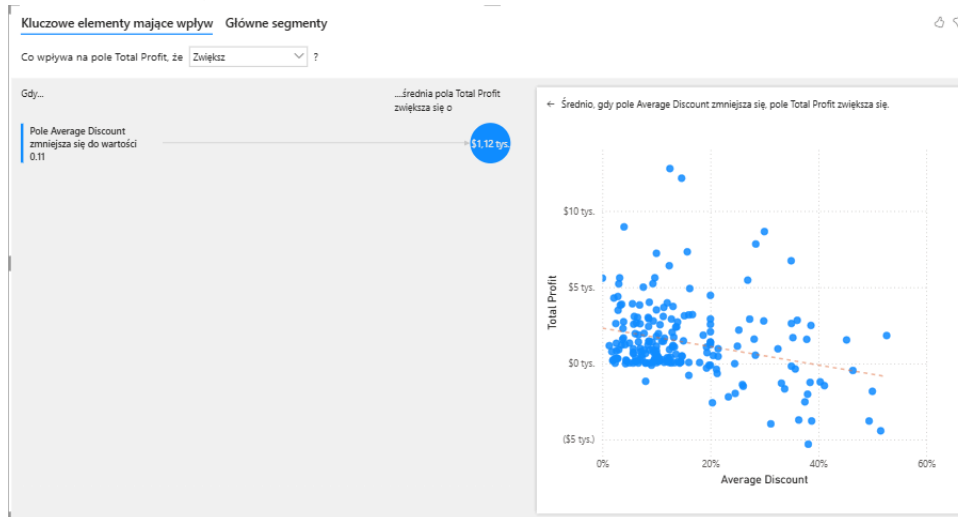
4. AI Features

Key Influencers

The Key Influencers visualization was used in the project to identify factors affecting the profit level. The analysis showed that one of the key factors increasing total profit is a lower discount level. As the average discount decreases, an increase in profit is observed, confirming the significant impact of discount policy on sales profitability.

When the Average Discount drops to approximately 11%, Total Profit increases by an average of approximately \$5.1K. Lower discounts mean higher profit — discounts have a significant impact on profitability.

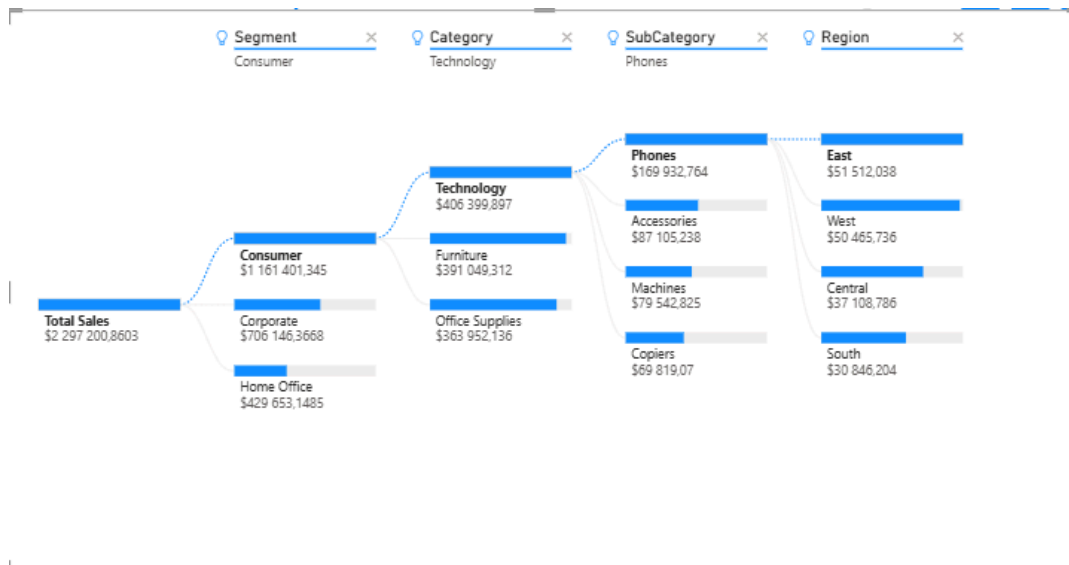
This is a sound, business-oriented conclusion.



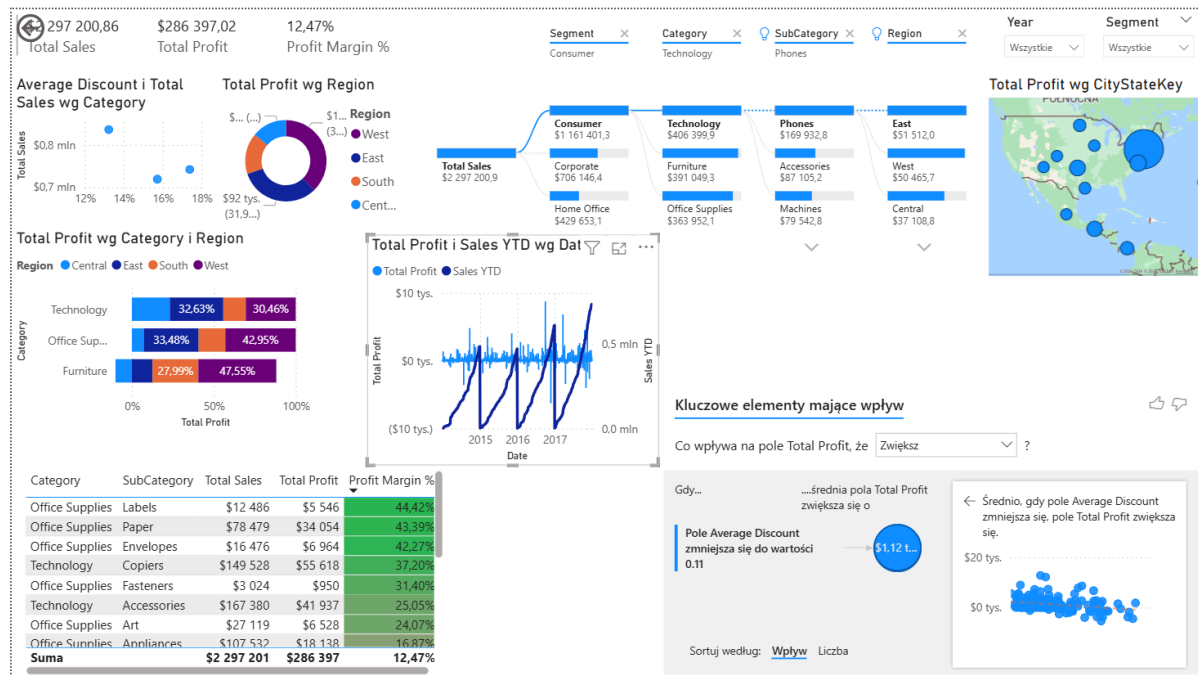
A further Key Influencers analysis showed that factors reducing profit levels include high discount levels and selected product subcategories, such as Supplies and Fasteners. An increase in the average discount leads to a significant decline in profit, indicating the need for careful management of discount policy.

Decomposition Tree

A decomposition tree was used to analyse the sales structure. The visualization enabled the identification of combinations of regions, categories, and segments generating the highest sales.

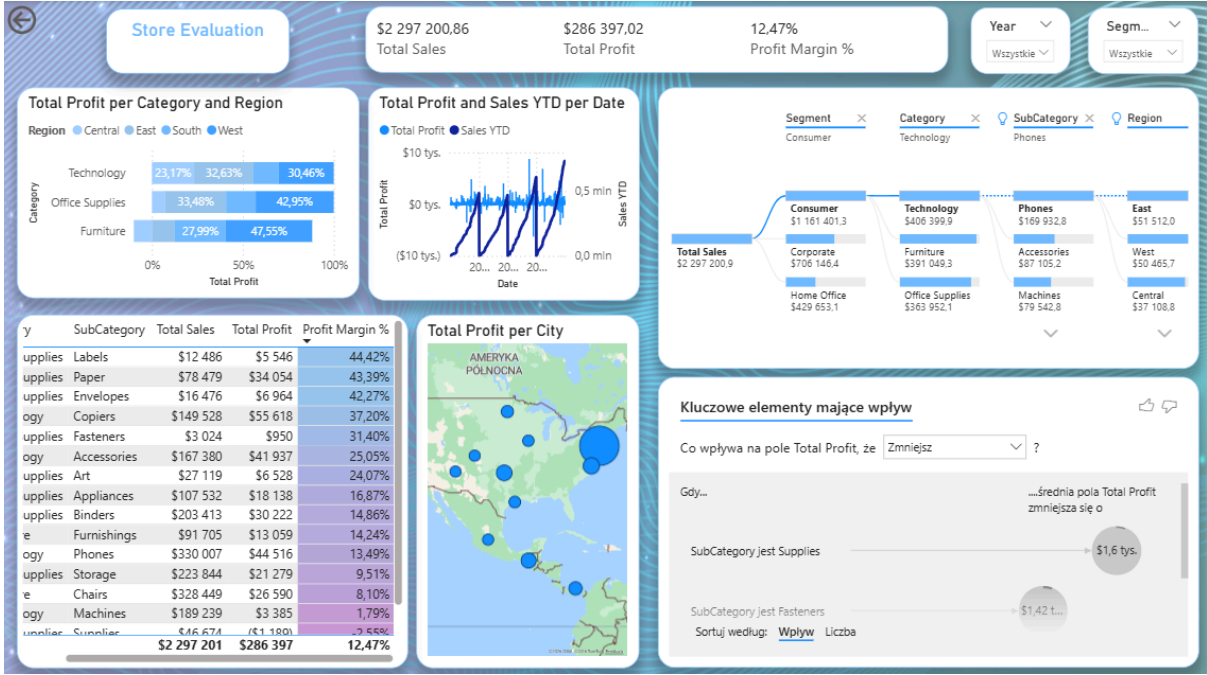


Dashboard before cleanup — with various test visualizations, before applying a consistent theme:



To prepare the final dashboard, a clear objective was defined: to answer "How do we sell, where, and why does profit change?" This ultimately led the project to the design shown

below.



Conclusions from Data Analysis and Visualization

Based on the data analysis conducted and the visualizations designed in Microsoft Power BI, the following conclusions were drawn:

Overall Sales and Profitability Results

Total sales value in the analysed period amounted to \$2,297,200.86, while total profit reached \$286,397.02, corresponding to a profit margin of 12.47%. This indicates the overall profitability of the company's operations; however, further analysis reveals significant differences between individual categories, regions, and product subcategories.

Sales and Profit Analysis Over Time

Time-series analysis of sales and profit showed clear fluctuations in values across individual periods, which is characteristic of transactional data. The Sales YTD (year-to-date sales) measure correctly resets at the beginning of each year, enabling comparison of sales dynamics between years. An overall upward trend in sales across consecutive years is visible, although profit levels do not always grow proportionally to sales.

Sales and Profit Structure by Segment and Category

The Consumer segment has the largest share of total sales, and the dominant product category is Technology, particularly the Phones subcategory. Analysis of profit structure by category and region showed that the West region generates the largest share of profit in most categories, while other regions have varying contributions depending on the product category.

Product Profitability Analysis

Detailed tabular analysis showed that not all high-sales products are simultaneously profitable. There are product subcategories, such as Tables and Bookcases, that generate negative profit and negative margin, meaning their sales result in losses. Conversely, subcategories such as Labels, Paper, and Envelopes are characterised by very high profit margins, despite relatively lower sales volumes.

Regional Analysis

The profit map by city shows a concentration of the highest profit values in the largest metropolitan areas, which is consistent with business expectations. This confirms the significant impact of location on the level of generated profit and the importance of high-customer-volume markets.

Conclusions from the Use of Artificial Intelligence (AI) Features

The use of the Key Influencers visualization enabled the identification of factors affecting changes in profit levels. The analysis showed that high discount levels significantly reduce profit, while limiting them leads to increased profitability. Additionally, selected product subcategories, such as Supplies and Fasteners, were identified as factors negatively

affecting profit. The AI analysis results are consistent with conclusions derived from classic data analysis.

Summary

The analysis conducted enabled a comprehensive assessment of the company's sales performance and profitability. Combining classic visualizations with artificial intelligence features made it possible not only to describe the data, but also to identify the causes of observed phenomena. The conclusions obtained may serve as a basis for further business decisions, particularly regarding discount policy and product range management.